MULTIPLEX COMMUNICATION SYSTEM

id100000001400

Outline

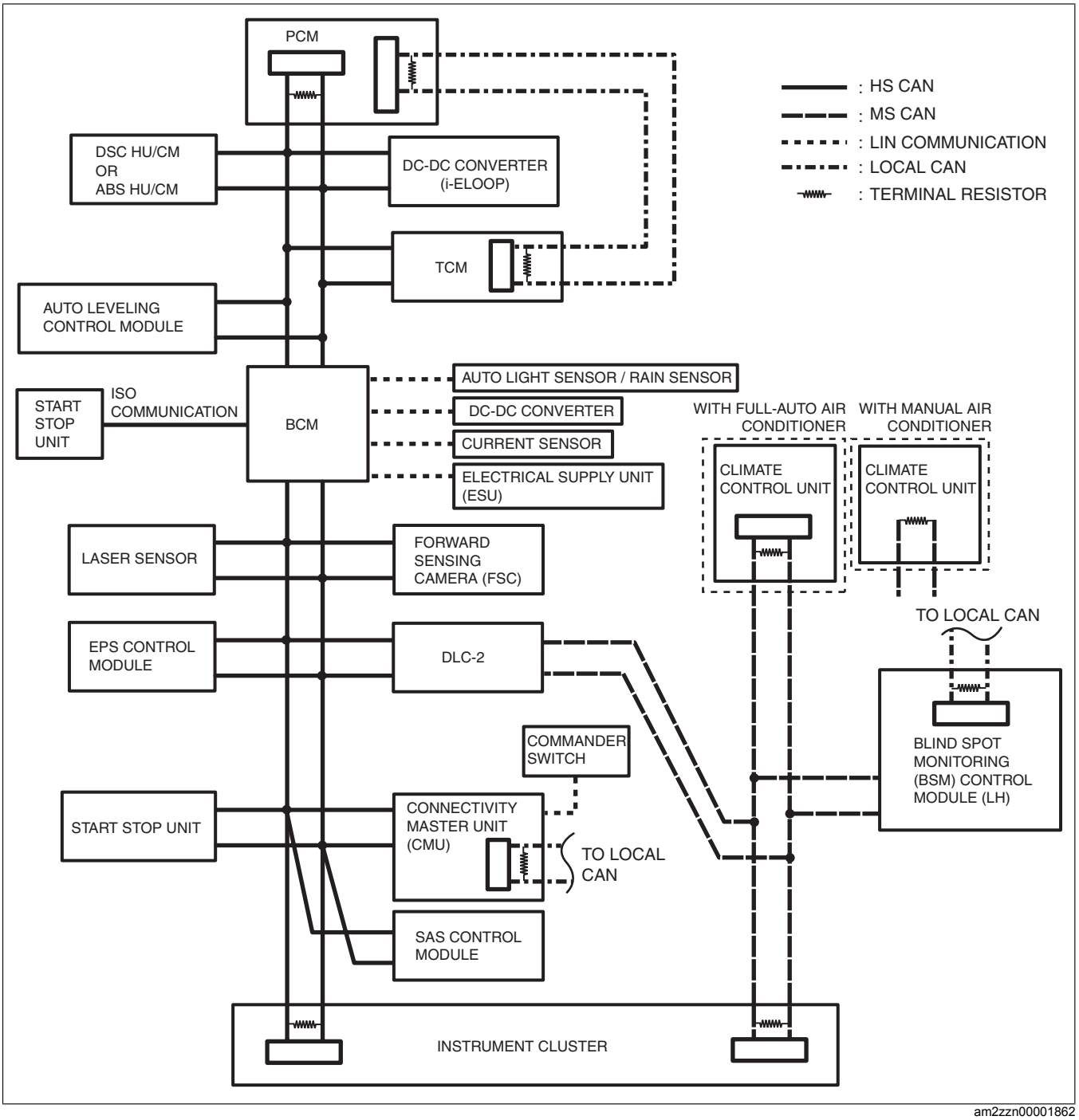
- A control area network (CAN) system has been adopted in which the multiple control modules send/receive signals using two common communication lines.
- Local CAN, ISO communication, or LIN communication has been adopted for the individual communication between control modules in consideration of the communication speed and the cost.

System Wiring Diagram

Note

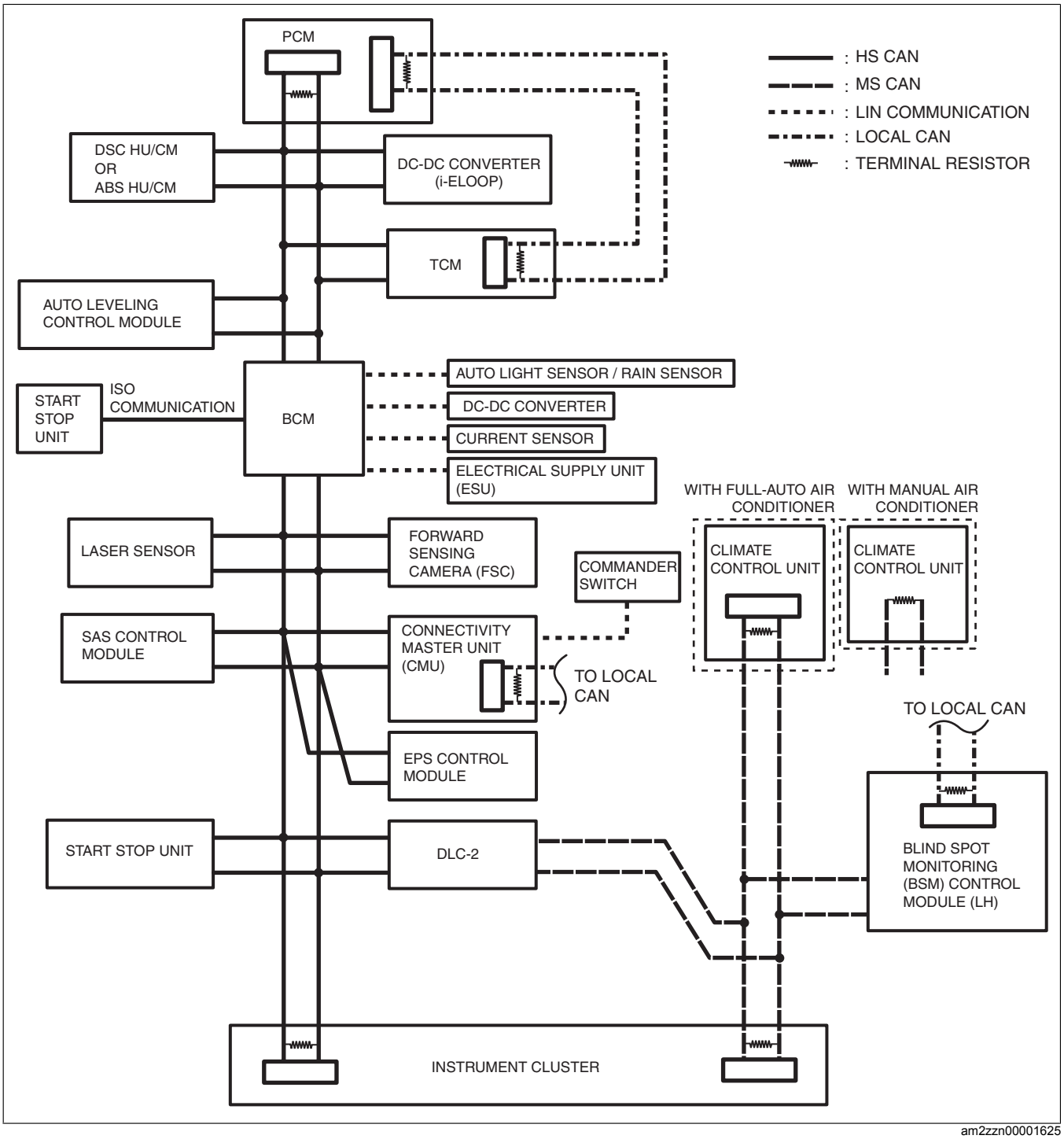
- The following figure shows the CAN/LIM communication/ISO communication/local CAN connection conditions. Availability depends on the vehicle specification.

CAN/LIN communication/ISO communication/local CAN (L.H.D. (SKYACTIV-G 1.5))



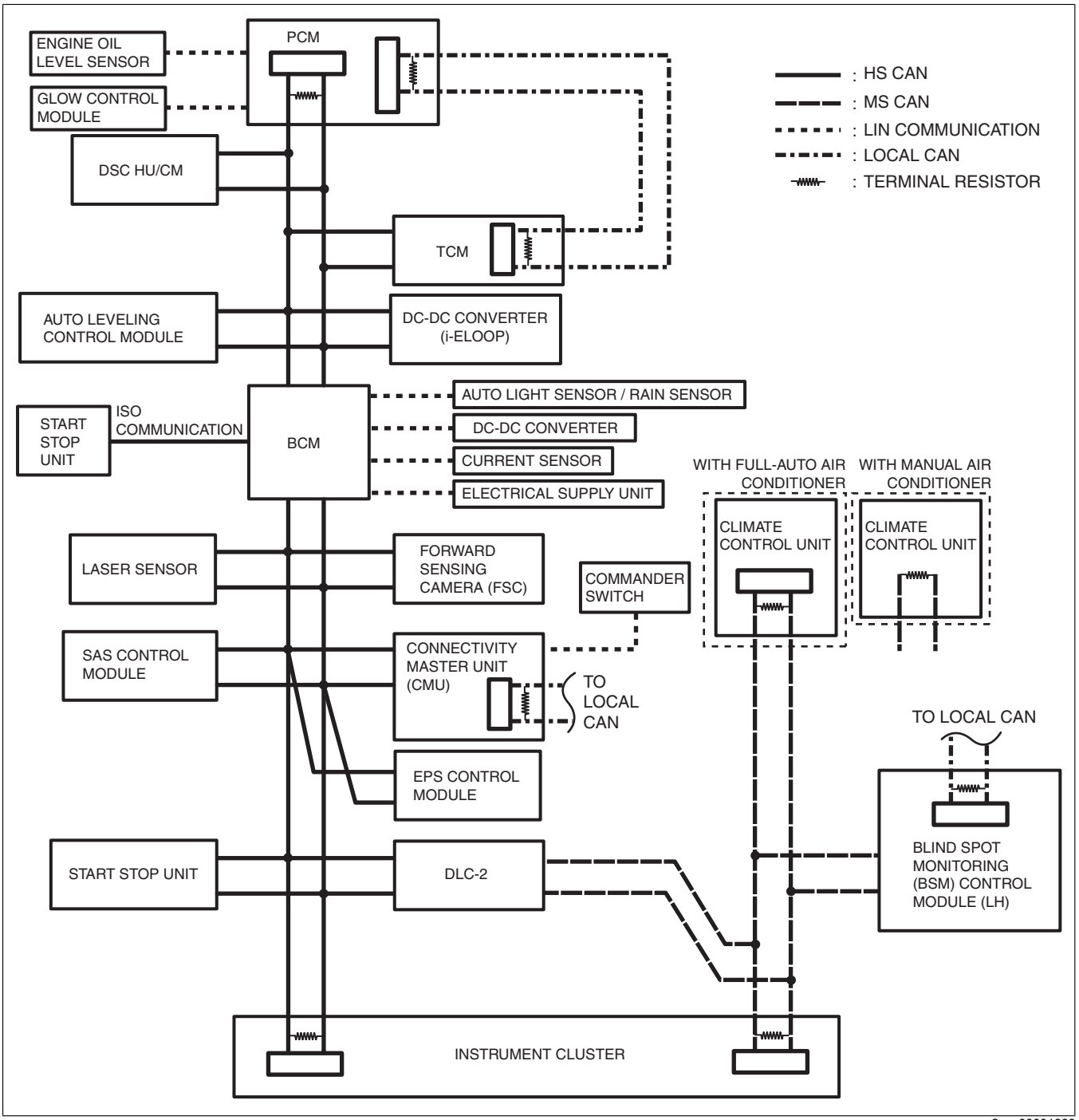
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CAN/LIN communication/ISO communication/local CAN (R.H.D. (SKYACTIV-G 1.3, SKYACTIV-G 1.5))



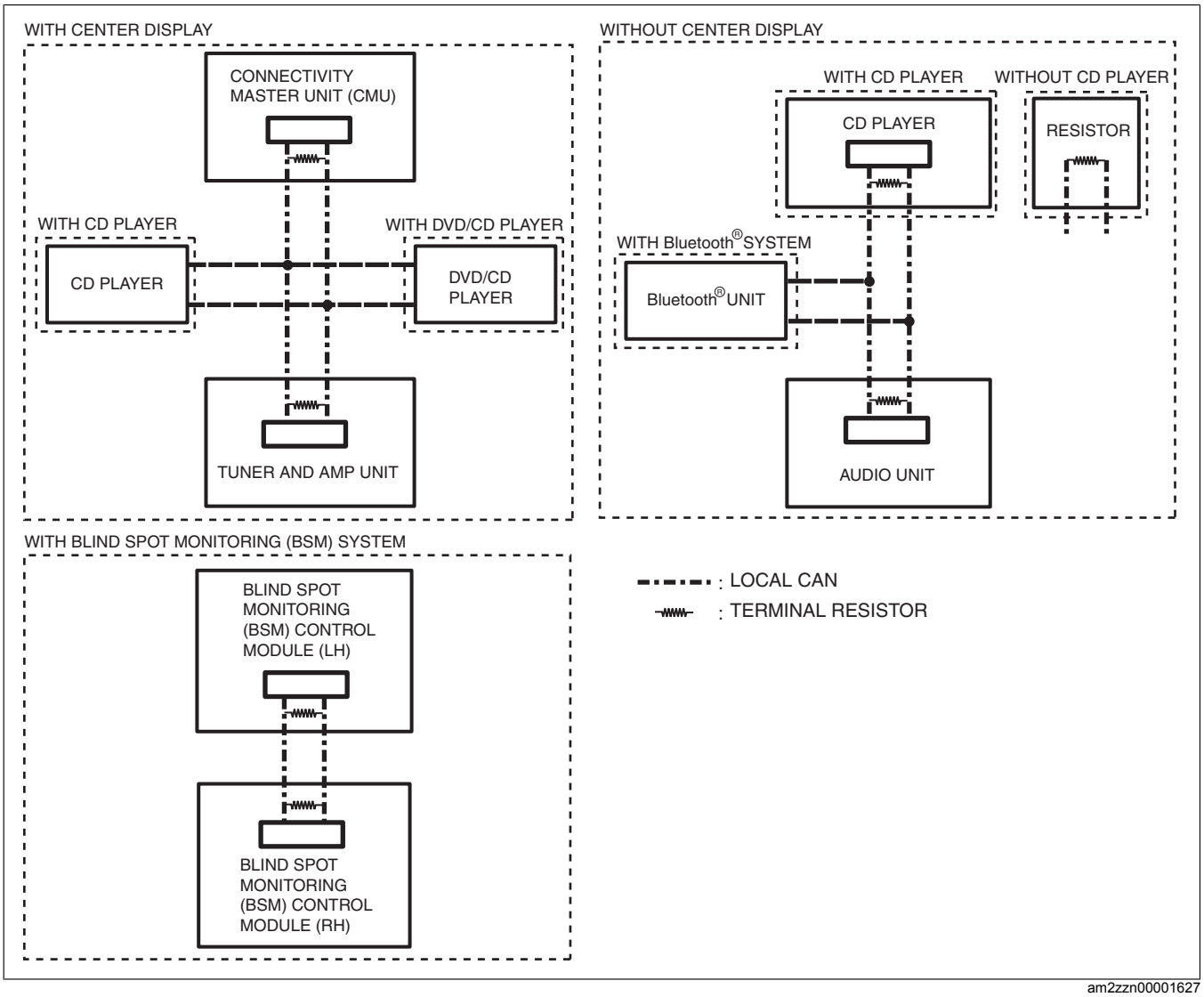
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CAN/LIN communication/ISO communication/local CAN (SKYACTIV-D 1.5)



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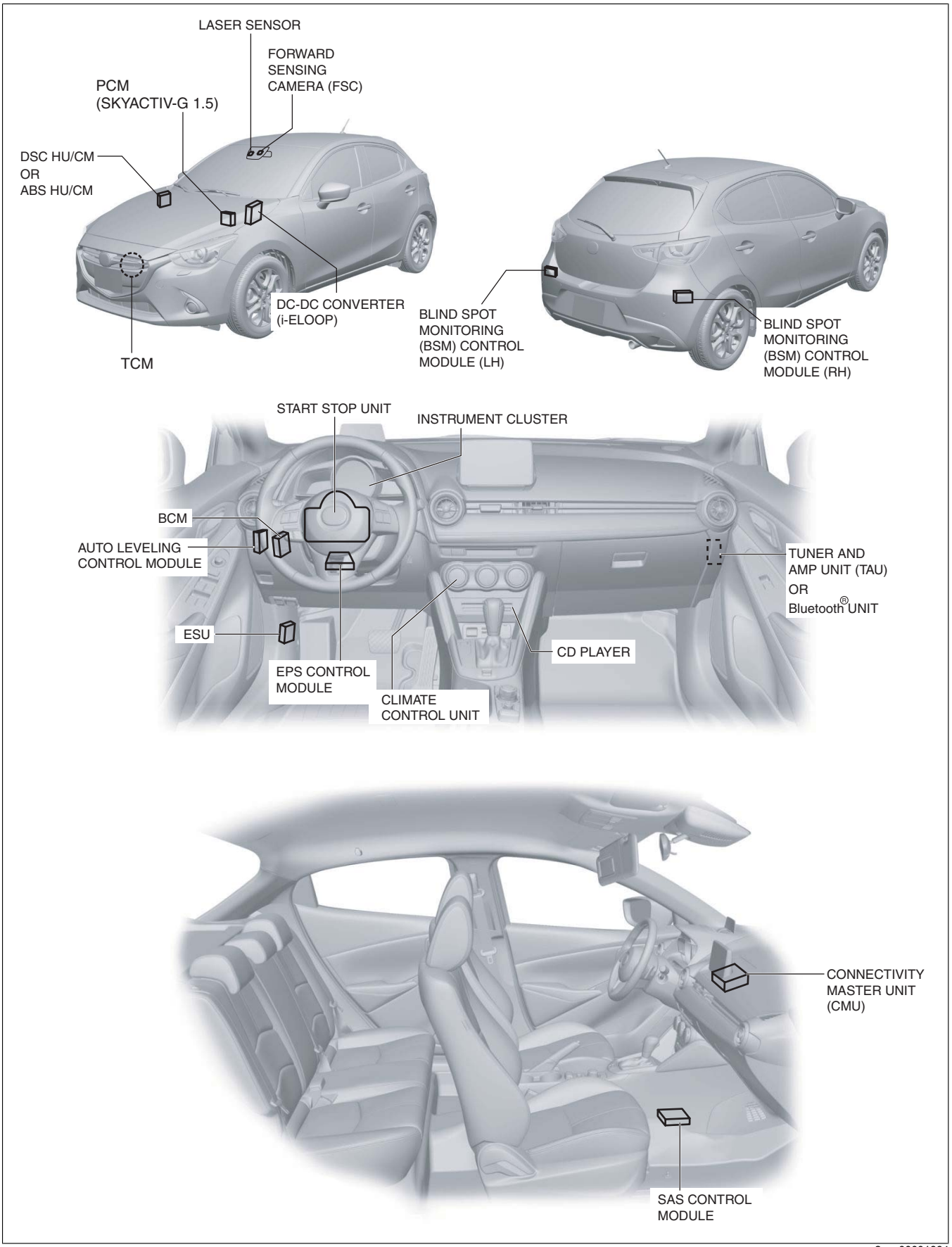
Local CAN



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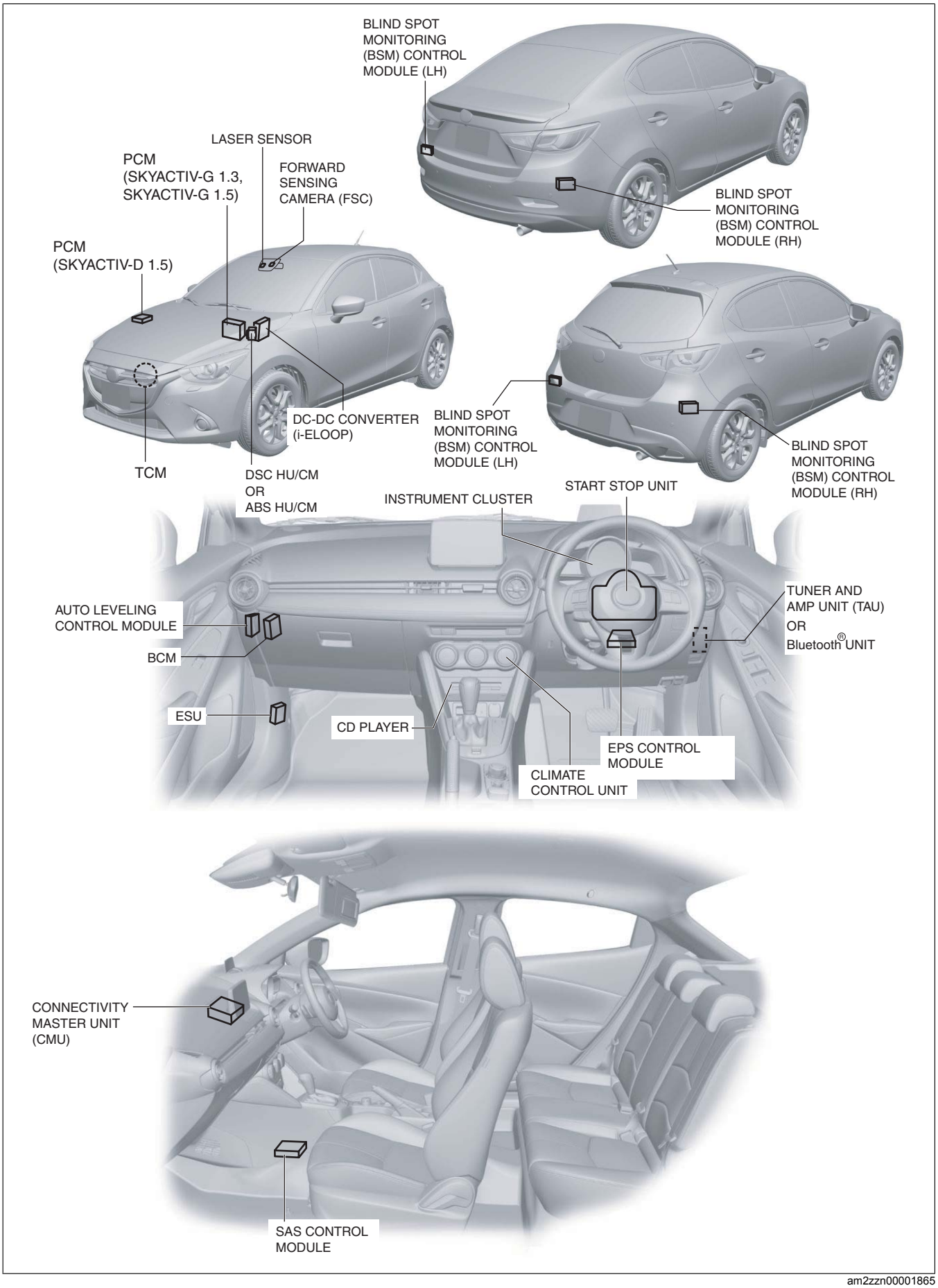
Structural view

L.H.D.



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R.H.D.



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Function

CAN (controller area network) system

- HS-CAN (high-speed controller area network) is used for communication between the following modules:
  - — PCM
  - — DSC HU/CM (with DSC system)
  - — ABS HU/CM (without DSC system)
  - — DC-DC converter (i-ELOOP) (With i-ELOOP system)
  - — TCM (ATX)
  - — Auto leveling control module (with headlight auto leveling system)
  - — BCM
  - — Laser sensor (with smart city brake support (SCBS))
  - — Forward sensing camera (FSC) (with forward sensing camera (FSC))
  - — EPS control module
  - — Start stop unit
  - — Connectivity master unit (CMU) (with center display)
  - — SAS control module
  - — Instrument cluster
- MS-CAN (mid-speed controller area network) is used for communication between the following modules:
  - — Instrument cluster
  - — Blind spot monitoring (BSM) control module (LH)
  - — Climate control unit (full-auto air conditioner)
  - — Climate control unit (Manual air conditioner)

*1 : The climate control unit for the vehicles equipped with the manual air conditioner system only uses the terminal resistance located on the substrate, therefore it does not use the CAN system.

- Signals transmitted using the CAN system are as follows:

HS-CAN communication table

OUT: Output (sends signal)
IN: Input (receives signal)

Signal	CAN system related module													
	PCM	**TCM**	**Auto leveling control module**	**Laser sensor**	**DSC HU/CM*2**	**ABS HU/CM*3**	**EPS control module**	**BCM**	**FSC**	**Instrument cluster**	**Start stop unit**	**SAS control module**	**DC-DC converter (i-ELOOP)**	**Connectivity master unit**
Acceleration status	–	–	–	–	IN	IN	–	–	–	–	–	OUT	–	–
Accelerator pedal opening angle information	OUT	–	–	–	IN	IN	–	–	IN	IN	–	IN	–	IN
A/C operation request	IN	–	–	–	–	–	–	–	–	OUT	–	–	–	IN
Air bag/front seat belt pre-tensioner system warning light on request	–	–	–	–	–	–	–	–	–	IN	–	OUT	–	IN
Air bag system-related information	–	–	–	–	–	–	–	–	–	IN	–	OUT	–	IN
Air bag system warning buzzer status	–	–	–	–	–	–	–	–	–	IN	–	OUT	–	IN
Ambient temperature	OUT	IN	–	IN	IN	–	–	IN	–	IN	–	IN	–	IN
	IN	IN	IN	IN	IN	–	IN	IN	–	OUT	IN	IN	–	IN
Back-up lights ON (MTX)	OUT	–	–	–	IN	–	–	–	IN	IN	–	IN	–	IN
Back-up light on request	–	–	IN	IN	IN	–	–	IN	–	OUT	–	–	–	IN

Signal	CAN system related module													
	PCM	TCM	Auto leveling control module	Laser sensor	DSC HU/ CM*2	ABS HU/ CM*3	EPS control module	BCM	FSC	Instrument cluster	Start stop unit	SAS control module	DC-DC converter (i-ELOOP)	Connectivity master unit
Battery discharge suppression status	–	–	–	–	–	–	–	OUT	–	IN	–	–	–	–
Battery regeneration status	OUT	–	–	–	–	–	–	–	–	IN	–	–	–	IN
BCM personalization-related information	–	–	–	–	–	–	–	IN	–	–	–	–	–	OUT
	–	–	–	–	–	–	–	OUT	–	–	–	–	–	IN
Blind spot monitoring (BSM) system personalization-related information	–	–	–	–	–	–	–	–	–	IN	–	–	–	OUT
	–	–	–	–	–	–	–	–	–	OUT	–	–	–	IN
Blind spot monitoring (BSM) system-related information	–	–	–	–	–	–	–	–	–	OUT	–	–	–	IN
	IN	–	–	–	–	–	–	OUT	–	–	–	–	–	–
Blower relay status	–	–	–	–	–	–	–	OUT	–	IN	–	–	–	–
Blower speed	–	–	–	–	–	–	–	–	–	OUT	–	–	–	IN
Brake fluid level	–	–	–	–	–	–	–	OUT	–	IN	–	–	–	IN
Brake light status	–	–	–	–	IN	–	–	OUT	–	IN	–	–	–	IN
Brake switch malfunction determination	OUT	–	–	–	–	–	–	–	–	IN	–	–	–	IN
Brake switch (No.1)	OUT	IN	–	–	IN	–	–	IN	IN	IN	–	–	–	IN
Brake switch (No.2)	OUT	IN	–	–	–	–	–	–	IN	IN	IN	IN	–	IN
Charging system warning light on request	OUT	–	–	–	–	–	–	IN	–	IN	–	–	–	IN
Check engine light on request	OUT	–	–	–	–	–	–	–	–	IN	–	–	–	IN
Cluster switch status	–	–	IN	–	–	–	–	–	IN	OUT	–	–	–	–
Clutch status	OUT	–	–	–	–	–	–	–	–	–	–	–	–	IN
Collision detection (front, side, roll over)	IN	–	–	–	IN	–	–	IN	–	–	–	OUT	–	IN
Combiner auto-dimming level	–	–	–	–	–	–	–	–	–	IN	–	–	–	OUT
Combiner dimming switching	–	–	–	–	–	–	–	–	–	IN	–	–	–	OUT
Combiner manual-dimming level	–	–	–	–	–	–	–	–	–	IN	–	–	–	OUT
Combiner open/close angle	–	–	–	–	–	–	–	–	–	IN	–	–	–	OUT
Cranking	IN	–	–	–	–	–	–	IN	–	IN	OUT	–	–	–

Signal	CAN system related module													
	PCM	TCM	Auto leveling control module	Laser sensor	DSC HU/ CM*2	ABS HU/ CM*3	EPS control module	BCM	FSC	Instrument cluster	Start stop unit	SAS control module	DC-DC converter (i-ELOOP)	Connectivity master unit
Cranking time	OUT	–	–	–	–	–	–	–	–	–	IN	–	–	IN
Cruise control switch	OUT	–	–	–	–	–	–	–	–	IN	IN	–	–	IN
	IN	–	–	–	–	–	–	–	–	OUT	–	–	–	IN
	IN	IN	–	–	–	–	–	–	–	–	OUT	–	–	–
DC-DC converter status	IN	–	–	–	–	–	–	OUT	–	–	–	–	–	–
Dimmer cancel	–	–	–	–	–	–	–	–	–	OUT	IN	–	–	IN
Door lock/ unlock status	–	–	–	–	–	–	–	–	–	–	OUT	–	–	IN
Door opening/ closing condition	IN	–	–	–	–	–	–	OUT	–	IN	IN	–	IN	IN
DPF indicator light on request	OUT*1	–	–	–	–	–	–	–	–	IN	–	–	–	IN
Driver's buckle switch status	IN	–	–	–	–	–	–	IN	–	IN	IN	OUT	–	IN
Driving condition	IN	–	IN	IN	IN	IN	IN	IN	–	OUT	IN	IN	–	IN
DSC system-related information	OUT	–	–	–	IN	–	–	–	–	–	–	–	–	–
	IN	IN	–	IN	OUT	–	IN	IN	IN	IN	–	IN	–	IN
DSC OFF switch status	–	–	–	–	IN	–	–	–	–	OUT	–	–	–	–
Electric AT oil pump-related (i-stop)	IN	OUT	–	–	–	–	–	–	–	–	–	–	–	–
Engine coolant temperature	OUT	–	–	–	–	–	–	IN	–	IN	–	–	–	IN
Engine displacement	OUT	IN	–	–	IN	–	IN	–	–	IN	–	–	–	IN
Engine rotation speed	OUT	–	–	–	IN	–	IN	IN	IN	IN	IN	IN	–	IN
Engine status	OUT	IN	–	–	IN	–	IN	IN	–	IN	IN	–	–	IN
Engine stop request	IN	–	–	–	–	–	–	–	–	OUT	–	–	–	IN
Engine switch status	–	–	IN	–	–	–	IN	IN	IN	OUT	–	–	–	IN
	IN	IN	–	–	–	–	–	IN	–	IN	OUT	–	–	IN
Engine torque	OUT	–	–	–	–	–	–	IN	–	–	–	–	–	–
EPS status	IN	–	–	–	–	–	OUT	–	–	IN	–	–	–	IN
Forward sensing camera (FSC) personalization-related information	–	–	–	–	–	–	–	–	IN	–	–	–	–	OUT
	–	–	–	–	–	–	–	–	OUT	–	–	–	–	IN
Front fog light information	–	–	–	–	–	–	–	OUT	IN	IN	–	–	–	IN
Front fog light switch position	–	–	–	–	–	–	–	IN	–	–	OUT	–	–	–
Fuel cut request	IN	–	–	–	IN	–	–	IN	–	IN	–	OUT	IN	IN
Fuel economy information reset request	–	–	–	–	–	–	–	–	–	IN	–	–	–	OUT

Signal	CAN system related module													
	PCM	TCM	Auto leveling control module	Laser sensor	DSC HU/ CM*2	ABS HU/ CM*3	EPS control module	BCM	FSC	Instrument cluster	Start stop unit	SAS control module	DC-DC converter (i-ELOOP)	Connectivity master unit
Fuel injection amount	OUT	–	–	–	–	–	–	–	–	IN	–	–	–	IN
Fuel-level sensor-related information	–	–	–	–	–	–	–	OUT	–	IN	–	–	–	–
Fuel tank level	IN	–	–	–	–	–	–	–	–	OUT	–	–	–	IN
Gear position	OUT	–	–	–	–	–	–	–	–	IN	–	IN	–	IN
	–	OUT	–	–	IN	IN	–	IN	IN	IN	IN	IN	–	IN
Global central configuration	IN	IN	IN	IN	IN	–	IN	IN	–	OUT	IN	IN	–	IN
Global central configuration error	OUT	–	–	–	–	–	–	–	–	IN	–	–	–	IN
Glow indicator light on request	OUT*1	–	–	–	–	–	–	–	–	IN	–	–	–	IN
Glow indicator light voltage	IN*1	–	–	–	–	–	–	–	–	OUT	–	–	–	–
Hazard warning switch information	–	–	–	–	–	–	–	IN	IN	IN	OUT	–	–	–
Headlight high-beam indicator light	–	–	–	–	–	–	–	OUT	IN	IN	–	–	–	IN
Headlight status	IN	–	IN	–	–	–	–	OUT	IN	IN	–	–	–	IN
Headlight warning request	–	–	–	–	–	–	–	OUT	–	IN	–	–	–	IN
High-beam control system personalization-related information	–	–	–	–	–	–	–	–	IN	–	–	–	–	OUT
	–	–	–	–	–	–	–	–	OUT	–	–	–	–	IN
High-beam on request	–	–	–	–	–	–	–	IN	OUT	–	–	–	–	–
Horn sound request	–	–	–	–	–	–	–	IN	–	–	OUT	–	–	–
i-ELOOP-related information	OUT	IN	–	–	–	–	–	–	–	IN	–	–	IN	IN
	IN	IN	–	–	–	–	–	IN	–	IN	–	–	OUT	IN
Immobilizer system-related information	OUT	–	–	–	–	–	–	–	–	–	IN	–	–	–
	IN	–	–	–	–	–	–	–	–	–	OUT	–	–	–
Instrument cluster personalization-related information	–	–	–	–	–	–	–	IN	IN	IN	IN	–	–	OUT
	–	–	–	–	–	–	–	–	–	–	OUT	–	–	IN
	–	–	–	–	–	–	–	–	OUT	–	–	–	–	IN
	–	–	–	–	–	–	–	OUT	–	–	–	–	–	IN
	–	–	–	–	–	–	–	–	–	OUT	–	–	–	IN
i-stop OFF switch status	IN	–	–	–	–	–	–	–	–	OUT	–	–	–	–
i-stop-related information	OUT	–	–	–	–	–	–	–	–	IN	–	–	–	IN
	IN	–	–	–	–	–	–	–	–	–	OUT	–	–	–
	IN	IN	–	–	OUT	–	–	–	–	–	–	–	–	–

Signal	CAN system related module													
	PCM	TCM	Auto leveling control module	Laser sensor	DSC HU/ CM*2	ABS HU/ CM*3	EPS control module	BCM	FSC	Instrument cluster	Start stop unit	SAS control module	DC-DC converter (i-ELOOP)	Connectivity master unit
i-stop warning light (amber)/i-stop indicator light (green) on request	OUT	IN	–	–	–	–	–	–	–	IN	IN	–	–	IN
Keyless warning alarm activation request	–	–	–	–	–	–	–	–	–	IN	OUT	–	–	IN
KEY warning light (amber)/ KEY indicator light (green) on request	–	–	–	–	–	–	–	OUT	–	IN	–	–	–	IN
	–	–	–	–	–	–	–	–	–	IN	OUT	–	–	IN
Laser sensor-related information	IN	–	–	OUT	–	–	–	–	–	–	–	–	–	IN
LED headlight warning light on request	–	–	OUT	–	–	–	–	–	–	IN	–	–	–	IN
Light switch information	–	–	–	–	–	–	–	IN	IN	IN	OUT	–	–	–
Maintenance monitor-related information	–	–	–	–	–	–	–	–	–	IN	–	–	–	OUT
Master warning light indication request	–	–	–	–	–	–	–	–	–	IN	–	–	–	OUT
Manual shift control	–	IN	–	–	–	–	–	–	–	OUT	–	–	–	–
	–	OUT	–	–	IN	IN	–	–	–	–	–	–	–	–
Manual shift inhibit flash command	–	OUT	–	–	–	–	–	–	–	IN	–	–	–	IN
Neutral switch	OUT	IN	–	–	–	–	–	–	–	IN	IN	IN	–	IN
Oil level warning	OUT	–	–	–	–	–	–	–	–	IN	–	–	–	IN
Oil level warning limit	OUT	–	–	–	–	–	–	–	–	–	–	–	–	IN
Oil warning light on request	OUT	–	–	–	–	–	–	–	–	IN	–	–	–	–
Panel light	–	–	–	–	–	–	–	–	–	OUT	IN	–	–	IN
Panel light on request	–	–	–	–	–	–	–	–	–	OUT	–	–	–	IN
Parking brake status	IN	IN	–	–	IN	IN	–	IN	–	OUT	–	–	–	IN
Passenger's buckle switch status	–	–	–	–	–	–	–	–	–	IN	–	OUT	–	IN
Passing switch information	–	–	–	–	–	–	–	IN	IN	IN	OUT	–	–	–
Power consumption	IN	–	–	–	–	–	OUT	–	–	–	–	–	–	IN
Prevention of brake override system unnecessary operation	OUT	–	–	–	–	–	–	IN	–	IN	–	–	–	IN

Signal	CAN system related module													
	PCM	TCM	Auto leveling control module	Laser sensor	DSC HU/ CM*2	ABS HU/ CM*3	EPS control module	BCM	FSC	Instrument cluster	Start stop unit	SAS control module	DC-DC converter (i-ELOOP)	Connectivity master unit
Priority recirculate request	OUT	–	–	–	–	–	–	–	–	IN	–	–	–	–
Rear fog indicator light information	–	–	–	–	–	–	–	OUT	–	IN	–	–	–	IN
Rear fog light switch position	–	–	–	–	–	–	–	IN	–	–	OUT	–	–	–
Rear washer switch position	–	–	–	–	–	–	–	IN	–	–	OUT	–	–	–
Rear window defogger information	IN	–	–	–	–	–	–	OUT	–	IN	–	–	–	IN
Rear window defogger operation request	–	–	–	–	–	–	–	IN	–	OUT	–	–	–	–
Rear wiper low speed status	–	–	–	–	–	–	–	OUT	IN	IN	–	–	–	IN
Rear wiper switch position	–	IN	–	–	–	–	–	IN	–	–	OUT	–	–	–
Rolling counter	–	–	–	–	IN	IN	–	–	–	–	–	OUT	–	–
ROOM fuse status	–	IN	–	–	–	–	–	–	IN	OUT	–	–	IN	–
	–	–	–	–	–	–	–	OUT	–	IN	–	–	–	IN
Seat belt status	–	–	–	–	–	–	–	OUT	–	IN	–	–	–	IN
Security indicator light on request	–	–	–	–	–	–	–	–	–	IN	OUT	–	–	IN
Selector lever position	–	OUT	–	–	–	–	–	–	–	IN	–	–	–	IN
	IN	OUT	–	–	–	–	–	–	–	–	–	–	–	–
Shift indicator light	OUT	–	–	–	–	–	–	–	–	IN	–	–	–	IN
Smart City Brake Support (SCBS)-related information	–	–	–	–	–	–	–	–	–	IN	–	–	–	OUT
Starter relay status	OUT	–	–	–	–	–	–	–	–	–	IN	–	–	–
Start stop unit personalization-related information	–	–	–	–	–	–	–	–	–	–	OUT	–	–	IN
	–	–	–	–	–	–	–	–	–	–	IN	–	–	OUT
Steering angle (absolute steering angle)	–	–	–	–	–	–	IN	–	–	–	OUT	–	–	–
Steering angle condition	IN	–	–	IN	–	–	OUT	–	–	–	–	–	–	IN
	IN	IN	IN	IN	IN	–	IN	IN	–	OUT	IN	IN	–	IN
Steering angle (estimated absolute steering angle)	IN	IN	IN	IN	IN	–	OUT	–	IN	IN	IN	IN	–	IN
Steering lock-related information	–	–	–	–	–	–	–	–	–	IN	OUT	–	–	IN

Signal	CAN system related module													
	PCM	TCM	Auto leveling control module	Laser sensor	DSC HU/ CM*2	ABS HU/ CM*3	EPS control module	BCM	FSC	Instrument cluster	Start stop unit	SAS control module	DC-DC converter (i-ELOOP)	Connectivity master unit
Steering shift switch	–	IN	–	–	–	–	–	–	–	–	OUT	–	–	–
Supply voltage	IN	–	IN	IN	IN	–	IN	IN	–	OUT	IN	IN	–	IN
Target gear position	–	OUT	–	–	IN	–	–	–	–	IN	–	–	–	IN
Temperature control dial status	IN	–	–	–	–	–	–	–	–	OUT	–	–	–	IN
TNS condition	–	–	–	–	–	–	–	OUT	–	IN	IN	–	–	IN
Transmission information	OUT	IN	–	–	–	–	–	–	–	IN	IN	–	–	–
Traveled distance	OUT	–	–	–	–	–	–	–	–	IN	–	–	–	IN
Tuner and amplifier unit (TAU) communication status	–	–	–	–	–	–	–	–	–	IN	–	–	–	OUT
Turn signal/ hazard warning light on request	–	–	–	–	–	–	–	OUT	IN	IN	–	–	–	IN
Turn switch position	–	IN	–	–	–	–	–	IN	IN	IN	OUT	–	–	–
Vehicle specification information	IN	IN	IN	IN	IN	–	IN	IN	–	OUT	IN	IN	–	IN
Vehicle speed	IN	–	–	IN	OUT	–	IN	–	IN	IN	IN	–	–	–
	OUT	–	IN	–	IN	IN	IN	IN	IN	IN	IN	IN	–	IN
Warning guidance-related information	–	–	–	–	–	–	–	–	–	IN	–	–	–	OUT
Washer level sensor status	–	–	–	–	–	–	–	OUT	–	IN	–	–	–	IN
Wheel speed	IN	IN	–	IN	OUT	–	IN	IN	IN	IN	IN	IN	–	IN
Windshield washer switch position	–	–	–	IN	–	–	–	IN	–	IN	OUT	–	–	–
Windshield wiper (INT) volume position	–	–	–	–	–	–	–	IN	–	–	OUT	–	–	–
Windshield wiper operation request for rain light sensor	–	–	–	–	–	–	–	OUT	IN	–	–	–	–	IN
Windshield wiper operation status	IN	–	–	IN	IN	–	–	OUT	IN	IN	–	–	–	IN
Windshield wiper switch position	–	IN	–	IN	–	–	–	IN	–	–	OUT	–	–	–
Yaw rate	–	–	–	–	IN	IN	–	–	–	–	–	OUT	–	–
	IN	IN	–	IN	OUT	–	IN	IN	IN	IN	–	–	–	IN

*1 : SKYACTIV-D 1.5
*2 : With DSC system
*3 : Without DSC system

MS-CAN communication table

OUT: Output (sends signal)
IN: Input (receives signal)

Signal	CAN system related module		
	Climate control unit	Instrument cluster	Blind spot monitoring (BSM) control module (LH)
A/C operation request	OUT	IN	–
Air bag/front seat belt pre-tensioner system warning light on request	IN	OUT	–
Ambient temperature	IN	OUT	IN
Back-up light on request	–	OUT	IN
Blind spot monitoring (BSM) system personalization-related information	–	OUT	IN
	–	IN	OUT
Blind spot monitoring (BSM) system-related information	–	IN	OUT
	–	OUT	IN
Blower relay status	IN	OUT	–
Blower speed	OUT	IN	–
Cluster switch status	–	OUT	IN
Cranking	IN	OUT	–
Dimmer cancel	IN	OUT	IN
Driver information system information	IN	OUT	–
Driving condition	IN	OUT	IN
Engine coolant temperature	IN	OUT	–
Engine displacement	IN	OUT	–
Engine rotation speed	IN	OUT	–
Engine status	IN	OUT	IN
Engine stop request	OUT	IN	–
Engine switch condition	IN	OUT	IN
Gear position	–	OUT	IN
Global central configuration	IN	OUT	IN
Instrument cluster personalization-related information	–	OUT	IN
	–	IN	OUT
Panel light	IN	OUT	–
Panel light on request	IN	OUT	IN
Priority recirculate request	IN	OUT	–
Rear window defogger information	IN	OUT	–
Rear window defogger operation request	OUT	IN	–
ROOM fuse status	IN	OUT	–
Steering angle condition	IN	OUT	IN
Steering angle (estimated absolute steering angle)	–	OUT	IN
Supply voltage	IN	OUT	IN
Temperature control dial status	OUT	IN	–
Traveled distance	IN	OUT	IN
Turn switch status	–	OUT	IN
Vehicle speed	IN	OUT	IN
Vehicle specification information	IN	OUT	IN
Windshield defroster	OUT	IN	–
Windshield wiper operation status	IN	OUT	–
Yaw rate	–	OUT	IN

Malfunction diagnosis procedure

- If a vehicle with a malfunction in a system controlled by a CAN system related module is brought in, verify the repair order form and the malfunctioning symptom first, then perform CAN malfunction diagnosis to determine if the malfunction cause is in the CAN system or not.
- For CAN malfunction diagnosis, the voltage at the CAN connection terminal on the DLC-2 is measured, and based on the measured value, the CAN circuit can be examined or the malfunction symptom can be determined.
- If the malfunction symptom is not an open circuit, inspect the voltage or continuity at the CAN circuit and determine the malfunctioning location.

- If the malfunction symptom is an open circuit, determine the area of the open circuit by using the displayed communication error DTC and the module in which communication has failed.

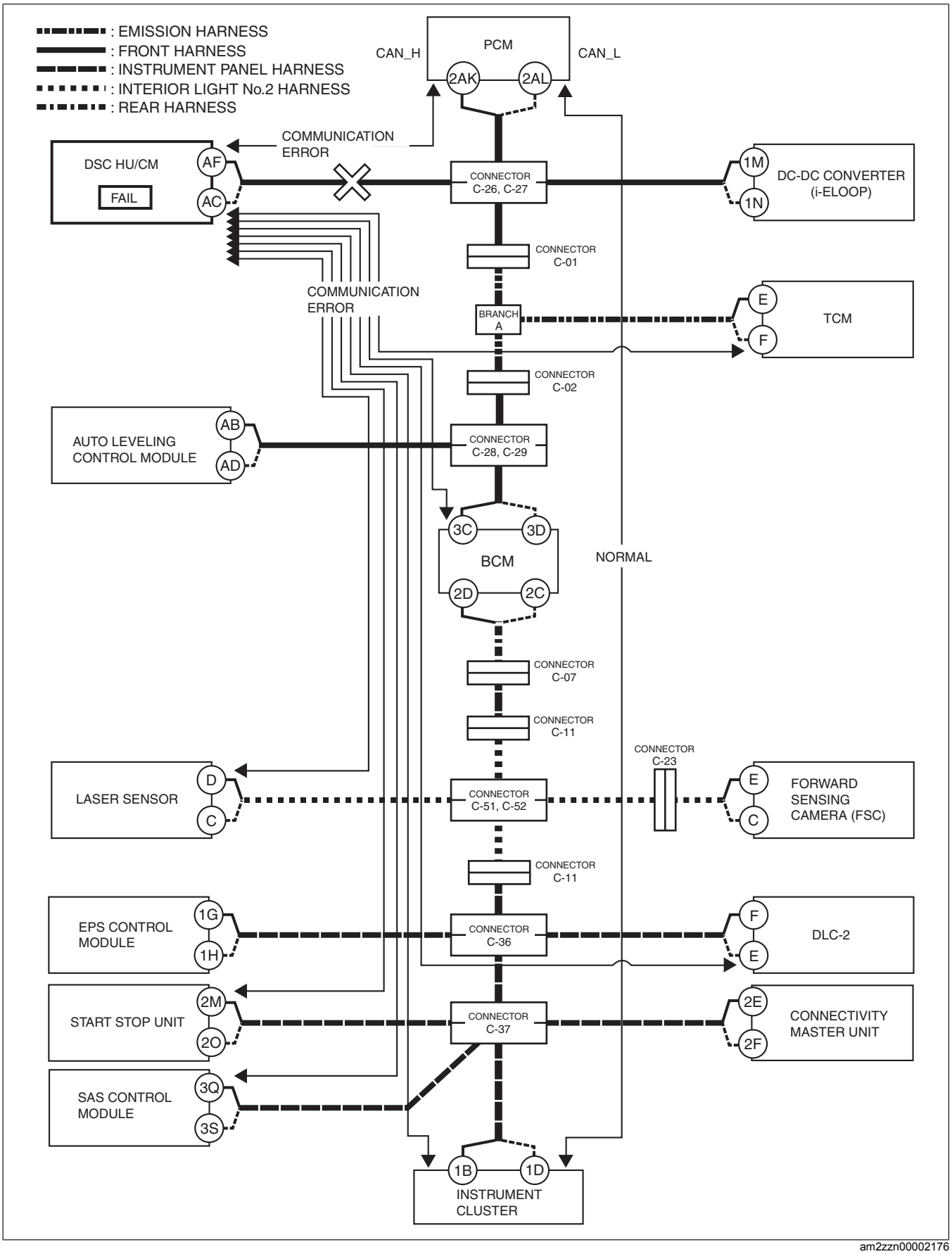
Ex.) Open circuit location determination procedure

1. Verify the CAN system-related module DTCs and the failed module using the Mazda Modular Diagnostic System (M-MDS).

DTC output module	Mazda Modular Diagnostic System (M-MDS) display	Output DTC
PCM	PCM	U0121:00
TCM	TCM	U0121:00
BCM	BCM	U0121:00
Laser sensor	SCBS	U0121:00
Forward sensing camera (FSC)	FSC	U0121:00
Connectivity master unit (CMU)	CMU	U0121:00
EPS control module	EPS	U0121:00
Start stop unit	SSU	U0121:00
Instrument cluster	IC	U0121:00

Module	Fail display
DSC HU/CM	×

2. As a result of DTC verification, only DTCs related to DSC HU/CM and communication errors are output and the DSC HU/CM is indicated as failed, therefore there could be a malfunction in the DSC HU/CM or in the wiring harness between connector C-26, C-27 and DSC HU/CM.



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Local CAN

- Local CAN is used for communication between the following modules:
  — Between PCM and TCM (ATX)

— Between connectivity master unit (CMU), DVD/CD player, CD player, and tuner and amp unit (with center display)
— Between audio unit, Bluetooth® unit, and CD player (without center display)
— Between blind spot monitoring (BSM) control module (LH) and blind spot monitoring (BSM) control module (RH) (with blind spot monitoring (BSM) system)

LIN communication
- LIN communication is used for communication between the following modules:
  — Between PCM and glow control module (SKYACTIVE-D 1.5)
  — Between PCM and engine oil level sensor (SKYACTIVE-D 1.5)
  — Between BCM and auto light sensor / rain sensor
  — Between BCM and DC-DC converter
  — Between BCM and current sensor
  — Between BCM and electrical supply unit
  — Between connectivity master unit (CMU) and commander switch

ISO communication
- ISO communication is used for communication between the following modules:
  — Between rear BCM and start stop unit

Construction
CAN
- The HS-CAN has terminator resistors built into the following units which form the CAN lines.
  — Between PCM (SKYACTIV-G 1.3, SKYACTIV-G 1.5) terminals 2AK (CAN_H) and 2AL (CAN_L), resistance value: **124 ohms**
  — Between PCM (SKYACTIV-D 1.5) terminals 2C (CAN_H) and 2D (CAN_L), resistance value: **124 ohms**
  — Between instrument cluster terminals 1B (CAN_H) and 1D (CAN_L), resistance value: **120 ohms**
- The MS-CAN has terminal resistors built into the following units which form the CAN lines.
  — Between climate control unit terminals D (CAN_H) and B (CAN_L), resistance value: **120 ohms** (with full-auto air conditioner)
  — Between climate control unit terminals W (CAN_H) and X (CAN_L), resistance value: **120 ohms** (with manual air conditioner)
  — Between instrument cluster terminals 1C (CAN_H) and 1E (CAN_L), resistance value: **120 ohms**

Local CAN
- The local CAN has terminator resistors built into the following units which form the CAN lines.
  — Between connectivity master unit (CMU) terminals 2I (CAN_H) and 2J (CAN_L), resistance value: **124 ohms** (with center display)
  — Between tuner and amp unit terminals 2O (CAN_H) and 2Q (CAN_L), resistance value: **120 ohms** (with center display)
  — Between audio unit terminals 1O (CAN_H) and 1Q (CAN_L), resistance value: **120 ohms** (without center display)
  — Between CD player terminals G (CAN_H) and H (CAN_L), resistance value: **120 ohms** (without center display)
  — Between blind spot monitoring (BSM) control module (LH) terminals G (CAN_H) and J (CAN_L), resistance value: **120 ohms** (with blind spot monitoring (BSM) system)
  — Between blind spot monitoring (BSM) control module (RH) terminals G (CAN_H) and J (CAN_L), resistance value: **120 ohms** (with blind spot monitoring (BSM) system)
  — Between PCM (SKYACTIV-G 1.3, SKYACTIV-G 1.5) terminals 1A (CAN_H) and 1B (CAN_L), resistance value: **120 ohms**
  — Between PCM (SKYACTIV-D 1.5) terminals 1G (CAN_H) and 1C (CAN_L), resistance value: **120 ohms**
  — Between TCM terminals G (CAN_H) and H (CAN_L), resistance value: **120 ohms**

LIN communication
- The LIN communication has drivers built into the following units which form the LIN communication lines.
  — PCM (SKYACTIV-D 1.5)
  — Glow control module (SKYACTIV-D 1.5)
  — Engine oil level sensor (SKYACTIV-D 1.5)
  — BCM
  — Auto light sensor / rain sensor
  — Electrical supply unit (ESU)
  — DC-DC converter
  — Current sensor
  — Connectivity master unit (CMU)
  — Commander switch

ISO communication

- The ISO communication has drivers built into the following units which form the ISO communication lines.
  - Start stop unit
  - BCM